

# MTH344 Sample Final (100 pts + EC 20 pts)

*Please print your name clearly. A penalty may be applied if your name is difficult to read.*

**Last Name:** \_\_\_\_\_

**First Name:** \_\_\_\_\_

## **Instructions:**

- There are nine problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

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**P1** (10 pts)

Let  $X_1, \dots, X_n$  be i.i.d. with PDF  $f(x; \theta)$ . Let  $T = t(X_1, \dots, X_n)$  be an unbiased estimator of  $\theta$ . Show that, under regularity conditions for the asymptotic normality of MLEs,

$$\text{Var}_\theta[T] \geq \frac{1}{\mathcal{I}_n(\theta)} \quad (1)$$

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**P2** (15 pts)

Let  $\mathbf{X}_i := (X_{i1}, \dots, X_{id})^T \stackrel{\text{i.i.d.}}{\sim} \text{Multin}(1, \mathbf{p})$  where  $i = 1, \dots, n$ . Show that:

$$\hat{\mathbf{p}}^{\text{MLE}} = (\bar{X}_{\cdot 1}, \dots, \bar{X}_{\cdot d})^T, \quad (2)$$

where  $\bar{X}_{\cdot j} = n^{-1} \sum_{i=1}^n X_{ij}$  and  $j = 1, \dots, d$ .

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**P3** (15 pts)

Let  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$ ,  $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$ , and  $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$ , where the parameter vector  $\theta = (\mu_X, \mu_Y, \sigma^2)^T$  is unknown,  $n \geq 2$ , and  $m \geq 2$ . Suppose we test  $H_0 : \mu_X = \mu_Y$  vs.  $H_1 : \mu_X \neq \mu_Y$

a. (5 pts) Show that the MLE on  $\Omega = \Omega_0 \cup \Omega_1$ ,  $\hat{\theta}^\Omega$ , is given by:

$$\hat{\theta}^\Omega = \left( \bar{X}, \bar{Y}, \frac{(n+m-2)S_p^2}{n+m} \right)^T, \quad (3)$$

where  $S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$ .

b. (5 pts) Show that the MLE on  $\Omega_0$ ,  $\hat{\theta}^{\Omega_0}$ , is given by  $\hat{\theta}^{\Omega_0} = \left( \hat{\mu}^{\Omega_0}, \hat{\mu}^{\Omega_0}, \hat{\sigma}^{2\Omega_0} \right)^T$ , where

$$\hat{\mu}^{\Omega_0} = \frac{n\bar{X} + m\bar{Y}}{n+m} \quad \text{and} \quad \hat{\sigma}^{2\Omega_0} = \hat{\sigma}^{2\Omega} + \frac{nm}{(n+m)^2} (\bar{X} - \bar{Y})^2. \quad (4)$$

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c. (5 pts) Since  $T \sim t(n + m - 2)$  under  $H_0 : \mu_X = \mu_Y$ , the GLR test of  $\alpha$  is given by  $d^{\text{GLR}}(\mathbf{X}, \mathbf{Y}) = I(|T| > t_{\alpha/2}(n + m - 2))$ . Let  $\pi_{d^{\text{GLR}}}(\theta) = E_{\theta}[d^{\text{GLR}}(\mathbf{X}, \mathbf{Y})]$  be the power function of the GLR test. Show that:

$$\begin{aligned} & \pi_{d^{\text{GLR}}}(\theta) \\ &= E_{\theta} \left[ 1 - \Phi \left( \frac{S_p t_{\alpha/2}(n + m - 2) - \frac{\mu_X - \mu_Y}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}}}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}} \right) + \Phi \left( -\frac{S_p t_{\alpha/2}(n + m - 2) - \frac{\mu_X - \mu_Y}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}}}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}} \right) \right]. \end{aligned} \tag{5}$$

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**P4** (10 pts)

Let  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$  with an unknown parameter vector  $\theta = (\mu, \sigma^2)^T$ , where  $n \geq 2$ . We test  $H_0 : \sigma \leq \sigma_0 (> 0)$  vs.  $H_1 : \sigma > \sigma_0$ .

a. (5 pts) Show that  $d^{\text{GLR}}(\mathbf{X}) = I\left(\frac{(n-1)S^2}{\sigma_0^2} > \chi_\alpha^2(n-1)\right)$  is the GLR test of level  $\alpha$ .

b. (5 pts) Show that  $d^{\text{GLR}}(\mathbf{X})$  is unbiased.

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## P5 (10 pts)

Suppose we design an experiment to test  $H_0 : \mu_X = \mu_Y$  vs  $H_1 : \mu_X \neq \mu_Y$ . For simplicity, assume  $\rho$  is known and variances are known and equal:  $\sigma_X^2 = \sigma_Y^2 = \sigma^2$ . Consider two experimental designs using a total of  $2n$  measurements ( $n \geq 2$ ):

- **Two Independent Samples:** Draw  $n$  independent subjects for  $X$  and  $n$  independent subjects for  $Y$ , i.e.,  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$  and  $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$ .
- **Paired Samples:** Draw  $n$  pairs  $(X_1, Y_1)^T, \dots, (X_n, Y_n)^T \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}\right)$ .

The GLR tests are equal-tail  $Z$ -tests. Show that the power for paired samples is larger than that for two independent samples if and only if  $\rho > 0$ .

## P6 (10 pts)

Let  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$  with an unknown parameter vector  $\theta = (\mu, \sigma^2)^T$ , where  $n \geq 2$ . We test  $H_0 : \sigma^2 = \sigma_0^2 (> 0)$  vs.  $H_1 : \sigma^2 \neq \sigma_0^2$ .

- a. (5 pts) Show that, for any  $0 < \lambda_0 \leq 1$ ,  $\Lambda(\mathbf{X}) < \lambda_0 \iff \frac{(n-1)S^2}{n\sigma_0^2} < c_L$  or  $\frac{(n-1)S^2}{n\sigma_0^2} > c_U$ , where the constants  $c_L$  and  $c_U$  satisfy  $c_L \leq c_U$  and  $\log \lambda_0 = \frac{n}{2}(\log c_L - c_L + 1) = \frac{n}{2}(\log c_U - c_U + 1)$ .

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b. (5 pts) For these hypotheses, any unbiased test  $d$  should satisfy  $\frac{\partial}{\partial \sigma^2} \pi_d(\theta)|_{\sigma^2=\sigma_0^2} = 0$ . Show that  $\frac{\partial}{\partial \sigma^2} \pi_{d^{\text{GLR}}}(\theta)|_{\sigma^2=\sigma_0^2} \neq 0$  when  $0 < \alpha < 1/2$ .



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**P7** (20 pts)

Let  $\phi(x)$  and  $\Phi(x)$  be the PDF and CDF of the standard Gaussian distribution, respectively.

a. (10 pts) *Mill's ratio* of the standard Gaussian is defined as  $m(x) = (1 - \Phi(x))/\phi(x)$ . For any  $x > 0$ , show that  $1 - \frac{1}{x^2} < xm(x) < 1$ .

b. (10 pts) Let  $g(x) = x\phi(x)$ . For any  $x > 0$ , show that  $g(x) \leq \frac{1}{\sqrt{2\pi}}e^{-1/2}$ .

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**P8** (10 pts)

We consider the following balanced two-way ANOVA model:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \text{ where } 1 \leq i \leq I, 1 \leq j \leq J, \text{ and } 1 \leq k \leq K \\ \sum_{i=1}^I \alpha_i = 0, \quad \sum_{j=1}^J \beta_j = 0, \quad \sum_{i=1}^I \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^J \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \end{cases}$$

where  $\theta = (\mu, \alpha_1, \dots, \alpha_{I-1}, \beta_1, \dots, \beta_{J-1}, \delta_{11}, \dots, \delta_{I-1, J-1}, \sigma^2)^T$  is unknown. Suppose we test:

$$H_0^A : \alpha_1 = \dots = \alpha_I = 0 \quad \text{vs.} \quad H_1^A : \text{Not } H_0^A. \quad (6)$$

a. (5 pts) Show that  $\text{SSA} \perp\!\!\!\perp \text{SSE}$ .

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b. (5 pts) Show that the MLE over  $\Omega_0$  (where  $\alpha_i = 0$ ) is given by:

$$\begin{aligned}\hat{\mu}^{\Omega_0} &= \bar{Y}_{...}, \quad \hat{\alpha}_i^{\Omega_0} = 0, \quad \hat{\beta}_j^{\Omega_0} = \bar{Y}_{.j.} - \bar{Y}_{...}, \\ \hat{\delta}_{ij}^{\Omega_0} &= \bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}, \quad \text{and} \quad \hat{\sigma}^{2\Omega_0} = \frac{\text{SSA} + \text{SSE}}{N}.\end{aligned}\tag{7}$$

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**P9** (EC 20 pts)

We consider the following balanced two-way ANOVA model:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \text{ where } 1 \leq i \leq I, 1 \leq j \leq J, \text{ and } 1 \leq k \leq K \\ \sum_{i=1}^I \alpha_i = 0, \quad \sum_{j=1}^J \beta_j = 0, \quad \sum_{i=1}^I \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^J \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \end{cases}$$

where  $\theta = (\mu, \alpha_1, \dots, \alpha_{I-1}, \beta_1, \dots, \beta_{J-1}, \delta_{11}, \dots, \delta_{I-1, J-1}, \sigma^2)^T$  is unknown. Suppose we test  $H_0^{AB} : \delta_{ij} = 0 \text{ for all } i, j$  vs.  $H_1^{AB} : \text{Not } H_0^{AB}$ .

a. (10 pts) Show that the MLE over  $\Omega = \Omega_0 \cup \Omega_1$  is given by:

$$\begin{aligned} \hat{\mu}^\Omega &= \bar{Y}_{...}, \quad \hat{\alpha}_i^\Omega = \bar{Y}_{i..} - \bar{Y}_{...}, \quad \hat{\beta}_j^\Omega = \bar{Y}_{.j.} - \bar{Y}_{...}, \\ \hat{\delta}_{ij}^\Omega &= \bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}, \quad \text{and} \quad \hat{\sigma}^2 = \frac{\text{SSE}}{N}. \end{aligned} \tag{8}$$

where  $N = IJK$ .

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b. (10 pts) Show that the MLE over  $\Omega_0$  (where  $\delta_{ij} = 0$ ) is given by:

$$\begin{aligned}\hat{\mu}^{\Omega_0} &= \bar{Y}_{...}, & \hat{\alpha}_i^{\Omega_0} &= \bar{Y}_{i..} - \bar{Y}_{...}, & \hat{\beta}_j^{\Omega_0} &= \bar{Y}_{.j.} - \bar{Y}_{...}, \\ \hat{\delta}_{ij}^{\Omega_0} &= 0, & \text{and } \hat{\sigma}^{2\Omega_0} &= \frac{\text{SSAB} + \text{SSE}}{N},\end{aligned}\tag{9}$$

where  $N = IJK$ .